



Important Formulas & Points to Remember: Operations with Integers

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1. Understanding Integers

Often, we are given the **sum** and **difference** of two numbers.

This helps us revise positive and negative numbers.

Example:

$$\text{Sum} = 25$$

$$\text{Difference} = 11$$

We try pairs:

(18, 7), (19, 6), (20, 5) etc.,

checking which pair satisfies both conditions.

Observation:

If you swap two numbers,

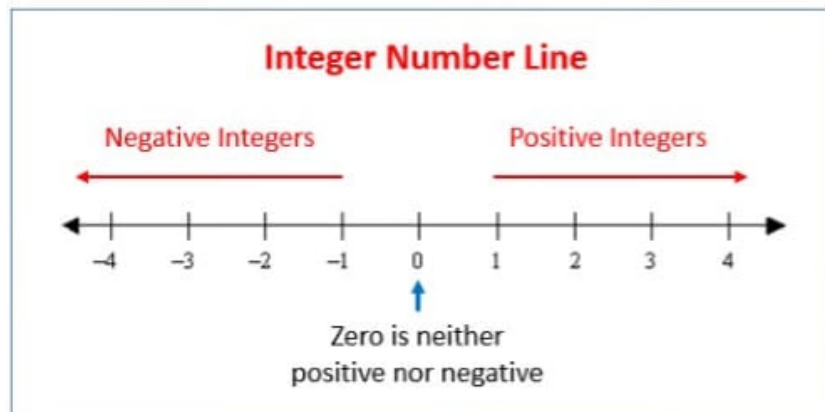
Sum stays same,

Difference changes sign.

This helps understand how positive and negative signs behave.

2. Number Line

Imagine a carrom coin or robot starting at **0**.



Moving **right** → positive integers

Moving **left** → negative integers

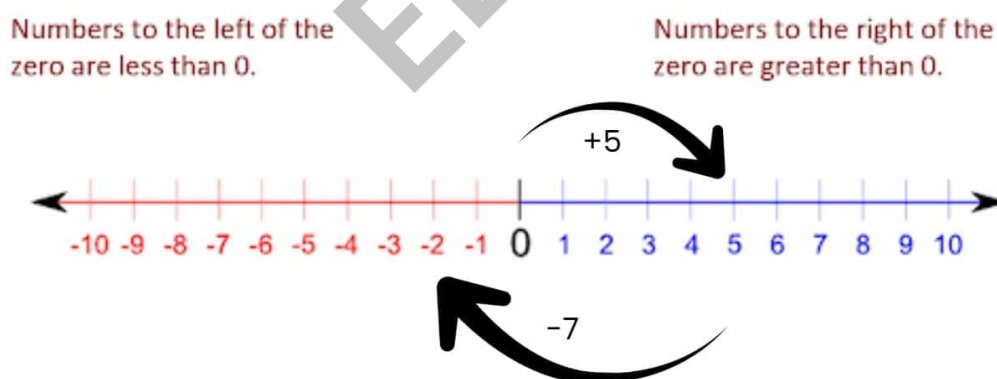
Example:

First movement = +5

Second movement = -7

Final position = $5 + (-7) = -2$

Meaning: the coin ends up 2 units left of zero.



3. Token Model

We imagine:

Green token = +1 (positive 1)

Red token = -1 (negative 1)

A **green + red together makes a zero pair**, because $+1 + (-1) = 0$.

Why is this helpful?

Because subtraction becomes simple:

If we want to do **7 - 18**,

we cannot remove 18 positives from 7.

So we add **11 zero pairs** (each pair adds +1 and -1 but total stays same).

This gives enough tokens to remove 18 positives.

Result = **-11**

Key takeaway:

Subtracting a number means:

Subtracting an integer = Adding its opposite
 $\boxed{\text{Subtracting an integer = Adding its opposite}}$

Subtracting an integer = Adding its opposite

Examples:

$$7 - 18 = 7 + (-18)$$

$$4 - (-12) = 4 + 12$$

This is an **essential rule** for all future algebra.

4. Additive Inverse

Every integer has an opposite:

Opposite of 5 is -5

Opposite of -12 is +12

Opposite of 0 is 0

Additive inverse of Additive inverse of a is -a

5. Multiplication of Integers

Multiplication is repeated addition -
but with integers, the **sign** plays a crucial role.

Let's break it into 4 cases:

CASE 1: Positive $\times$ Positive

Example: 4×2

Meaning: Add 2 four times $\rightarrow +8$

$$+ \times + = +$$

CASE 2: Positive $\times$ Negative

Example: $4 \times (-2)$

Meaning: Add -2 four times $\rightarrow -8$

$$+ \times - = -$$

CASE 3: Negative $\times$ Positive

Example: -4×2

Meaning: Remove +2 four times $\rightarrow$ total becomes -8

$$- \times + = -$$

CASE 4: Negative $\times$ Negative

Example: $(-4) \times (-2)$

Meaning: Remove -2 four times $\rightarrow$ leaves +8

$$- \times - = +$$

Shortcut:

Same signs $\rightarrow$ positive answer

Different signs $\rightarrow$ negative answer

6. Important Properties of Integer Multiplication

These properties will be used in algebra, equations, and simplifications.

1. Commutative Property

Order does not matter.

$$a \times b = b \times a$$

Examples:

$$3 \times -4 = -12$$

$$-4 \times 3 = -12$$

2. Associative Property

Grouping does not matter.

$$a \times (b \times c) = (a \times b) \times c \quad a \times (b \times c) = (a \times b) \times c$$

Example:

$$5 \times (-3 \times 4) = (5 \times -3) \times 4 = -60$$

3. Distributive Property

Important for simplification:

$$a(b + c) = ab + ac \quad a(b+c) = ab + ac$$

Example:

$$5 \times (4 + -2) = 5 \times 4 + 5 \times (-2)$$

$$= 20 - 10$$

$$= 10$$

4. Multiplying by 1

$$1 \times a = a$$

5. Multiplying by -1

$$-1 \times a = -a \quad -1 \times a = -a$$

Meaning: multiply by -1 flips the sign.

7. Division of Integers

Division follows the *same sign rules* as multiplication.

Shortcut:

Same signs → positive

Different signs → negative

8. Determining the Sign of a Long Product

Very helpful trick:

Count number of negative terms

If **even number** of negatives → product **positive**

If **odd number** of negatives → product **negative**

Example:

$$(-1) \times (-1) \times (-1) \times (-1)$$

4 negatives → positive

$$(-1) \times (-1) \times (-1)$$

3 negatives → negative

